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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: COMPILER DESIGN (24CS2235F) A.Y. 2026 - 2027

TITLE OF THE PROJECT	Syntax Error Detection and Correction Tool	
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Abstract

The **Syntax Error Detection and Correction Tool** is a compiler-based application designed to detect, analyse, and suggest corrections for syntax errors in source code. Syntax errors such as missing semicolons, unmatched brackets, incorrect keywords, and invalid expressions can make program development difficult, especially for beginners. This project provides an automated way to identify such errors and display clear error messages with possible corrections.

The system follows important **Compiler Design** concepts, including **lexical analysis, tokenization, syntax analysis, parsing, and error recovery**. The lexical analyser converts the source code into tokens, which are then passed to the syntax analyser. The parser checks these tokens against the grammar rules of the programming language and identifies invalid syntax. When an error is detected, the tool displays its location, type, and a possible correction. Basic error recovery techniques are also used to continue analysing the remaining code.

The main objective of this project is to provide a simple and user-friendly tool that demonstrates how compilers detect and handle syntax errors. It helps users understand grammar, parsing, and error recovery while reducing the time required to identify and correct common programming errors.

Signature of the Guide HOD